

Liam Imagawa

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EDUCATION

University of California, Berkeley

B.S. Aerospace Engineering (Dynamics & Control) — Minor in Physics (Quantum / Solid-State) Expected May 2027

M.S. Aerospace Engineering — Cumulative GPA: 3.58 / Major GPA: 3.81 Expected 2028

Dynamics & Control: Signals & Systems (EE 120), Dynamics (ME 104), Feedback Control (ME 132), Autonomous Flight (ME 136), Analytical Mechanics (PHYS 105)

Quantum & Math: Quantum Mechanics (PHYS 137A/B), Solid State (PHYS 141A), C^* -Algebras (MATH 208)

WORK EXPERIENCE

Berkeley AI Research (BAIR) Lab — PI: Prof. Murat Arcak

Sep 2025 – Present

NSF IRES Program (NTNU Norway Collaboration):

Sep 2025 – June 2026

- Designed decentralized multi-agent informative path planning algorithms for Autonomous Surface Vessels (ASVs), targeting algae bloom detection in the Arctic to enhance human safety and ocean awareness.
- Combined multi-agent RL with IPOPT-based MPC to plan paths robust to wind and current disturbances, improving planning iteration time to minutes while preserving MPC stability.
- Developed a localized data pipeline integrating Neptus infrastructure and SSH protocols for the active control, telemetry monitoring, and multi-stream visualization of ASV operational missions.

Collision Risk Metric Research (WIP, targeting ACC 2026)

github.com/LiamI16/risk-metrics

- Derived closed-form equivalences unifying scalar CPA metrics (TCPA/DCPA) with set-based Velocity Obstacles across finite and infinite horizons, connecting two previously disjoint collision-risk representations.
- Extended VOs to bounded convex uncertainty via a support-function representation (reducing Minkowski sums to point-wise addition) with complexity results, and built a linear-time VO library for arbitrary convex geometries.

NASA Ames Research Center Summer Intern — Aerospace Software Engineering

May 2025 – Aug 2025

- Architected a GMAT-based orbit propagation pipeline converting high-level mission parameters into automated trajectory analysis (state vectors, eclipse timings, perturbations) for downstream subsystem teams (GNC, power, comms), replacing a manual workflow.
- Built a feedforward neural-network model (PyTorch) that suggests near-optimal orbital elements for a target area of interest, replacing slow propagation with ML inference to cut optimization from hours to minutes (~ 0.88 Spearman correlation to ground truth).

EE C128 Instructional Lab Technician

Aug 2026 – Dec 2026

Design, dry-run, and debug hands-on lab experiments for a feedback-control course; validate hardware/software setups before student deployment, and support students during lab sessions later in the semester

PROJECTS

Quantitative Developer — Agentic Prediction-Market Trading System

Python, Asyncio, RAG, LLM Agents

- Led a 3-person team building a high-speed, client-agnostic trading engine for real-time L2 order-book processing.
- Developed a market-making framework using Hawkes processes and Unscented Kalman Filters for high-frequency volatility estimation and order-flow-imbalance modeling.
- Built a strategy-retrieval RAG database and an agentic backtesting engine to evaluate candidate strategies against historical prediction-market data.

Autonomous Flight Hardware Lab (ME 136)

C++, Real-Time Control

- Implemented a real-time flight controller in C++, processing raw accelerometer and IMU data to estimate vehicle attitude and stabilize autonomous flight.

Loam — Offline Generative Audio Engine

github.com/LiamI16/loam — TypeScript, Web Audio

- Architected and shipped a real-time generative audio engine with a framework-agnostic, unit-testable core (pure event-emitting logic, zero audio dependency) behind a thin synthesis adapter; deployed solo as a dependency-free offline PWA.
- Implemented the generation pipeline from stochastic primitives: a seeded PRNG and fractional Brownian motion shaped to a $1/f$ (pink) spectrum, driving Markov-chain harmonic progression under real-time scheduling.

TECHNICAL SKILLS

- GNC & Estimation:** MPC, Kalman Filtering, HMM, PID, State-Space, Fourier Analysis.
- Machine Learning:** CUDA, Neural Networks (CNN/MLP), Stable-Baselines3, PyTorch, TensorFlow, Gaussian Processes, Reinforcement Learning (MARL).
- Software:** Python (Asyncio, NumPy, SciPy), Java, C#, MATLAB/Simulink, Git, GMAT, LaTeX.